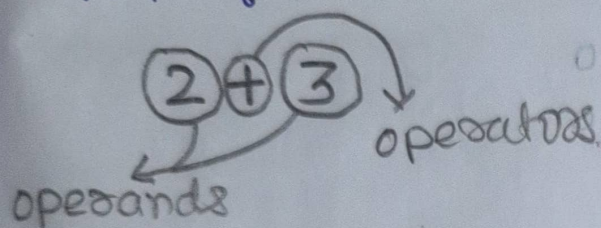


Operators

Operators
⇒ Types
⇒ Precedence

Unicode
→ ASCII

⇒ Operators are special symbols in Python to perform operations on operands.



Types of operators

- ⇒ Arithmetic operators
- ⇒ Assignment operators
- ⇒ Logical operators
- ⇒ Bitwise operators
- ⇒ Comparison operators
- ⇒ Membership operators
- ⇒ Identity operators

Arithmetic Operators

- + → Addition
- → Subtraction
- * → Multiplication
- / → Division
- % → Modulus (Remainder)
- ** → Exponent
- // → Floor division

a = 10, b = 5

$$\begin{aligned}
 a + b &= 15 \\
 a - b &= 5 \\
 a * b &= 50 \\
 a / b &= 2.0 \text{ (Float)} \\
 a \% b &= 0 \\
 a ** b &\Rightarrow a^b \Rightarrow 10^5 \Rightarrow 100000 \\
 a // b &\Rightarrow \lfloor a/b \rfloor \Rightarrow 2 \\
 &\Rightarrow 3 // 2 \Rightarrow \lfloor 3/2 \rfloor = 1
 \end{aligned}$$

$$\begin{aligned}
 a &= -12 // 5 \\
 \text{print}(a) &\Rightarrow -3 \\
 \lfloor -12/5 \rfloor & \\
 \lfloor -2.4 \rfloor &
 \end{aligned}$$

$$\begin{aligned}
 a &= 12 \% 5 \\
 \text{print}(a) &= 3 \\
 &\Rightarrow 12 \% 5 \\
 &\Rightarrow (-18 + 3) \% 5 \\
 &\Rightarrow 3
 \end{aligned}$$

$$\begin{aligned}
 &-22 \% 8 \\
 &(-24 + 2) \% 8 \\
 &= 0
 \end{aligned}$$

Comparison Operator

a=0, b=5

$== \rightarrow \text{print}(a==b) \Rightarrow \text{False}$
 $!= \rightarrow \text{print}(a!=b) \Rightarrow \text{True}$
 $<= \rightarrow \text{print}(a<=b) \Rightarrow \text{False}$
 $>= \rightarrow \text{print}(a>=b) \Rightarrow \text{True}$
 $< \rightarrow \text{print}(a<b) \Rightarrow \text{False}$
 $> \rightarrow \text{print}(a>b) \Rightarrow \text{True}$

Assignment Operator

$= \quad a=5$
 $+= \quad a+=7 \Rightarrow a=a+7=5+7=12$
 $-= \quad a-=2 \Rightarrow a=a-2=12-2=10$
 $*= \quad a*=2 \Rightarrow a=a*2=10*2=20$
 $/= \quad a/=5 \Rightarrow a=a/5=20/5=4.0$
 $\%= \quad a\%=3 \Rightarrow a=a\%3=4\%3=1.0$
 $**= \quad a**=2 \Rightarrow a=a**2=1**2=1.0$
 $//= \quad a//=2 \Rightarrow a=a//2=1//2=0$

Logical Operators

AND

a	b	a AND b
False	False	False
False	True	False
True	False	False
True	True	True

a	b	c	a AND b AND c
True	True	True	True
True	True	False	False
T	F	T	F

$\rightarrow$ Short circuiting

$\Rightarrow$ If any of the value becomes false then python doesn't check any other value. It returns simply false only.

x=4, y=10

$\text{print}(x>4 \text{ and } y>8) \Rightarrow \text{True}$
 $\text{print}(x>4 \text{ and } y<8) \Rightarrow \text{False}$
 $\text{print}(x<4 \text{ and } y>8) \Rightarrow \text{False}$
 $\text{print}(x<4 \text{ and } y<8) \Rightarrow \text{False}$

$\bullet \text{print}(3 \text{ and } 0 \text{ and } 4)$

$\Rightarrow \text{O/P} \Rightarrow 0$
False $\leftarrow$ last false value is printed

$\bullet \text{print}(2 \text{ and } 5 \text{ and } 7)$

$\Rightarrow \text{O/P} \Rightarrow 7$
True $\leftarrow$ last true value is printed.

OR

$$x=5, y=10$$

a	b	a OR b
False	False	False
True	False	True
False	True	True
True	True	True

$\text{print}(x > 4 \text{ OR } y > 8) \Rightarrow \text{True}$
 $\text{print}(x > 4 \text{ OR } y < 8) \Rightarrow \text{True}$
 $\text{print}(x < 4 \text{ OR } y > 8) \Rightarrow \text{True}$
 $\text{print}(x < 4 \text{ OR } y < 8) \Rightarrow \text{False}$

a	b	c	a OR b OR c
False	False	False	False
False	False	True	True
False	True	...	True
True	...	...	True

(Short Circuiting)

$\text{print}(2 \text{ OR } 3 \text{ OR } 5) \Rightarrow \text{True}$
 $\text{O/P} \Rightarrow 2$
 $\text{print}(0 \text{ OR } 7 \text{ OR } 9) \Rightarrow \text{True}$
 $\text{O/P} \Rightarrow 7$
 $\text{print}(0 \text{ OR } 0 \text{ OR } 11) \Rightarrow \text{True}$
 $\text{O/P} \Rightarrow 11$

NOT

$\text{print}(\text{NOT}(\text{True})) \Rightarrow \text{False}$
 $\text{print}(\text{NOT}(\text{False})) \Rightarrow \text{True}$
 $\text{print}(\text{NOT}(2 \text{ and } 0 \text{ and } 5)) \Rightarrow \text{True}$

Bitwise Operators

	a	b	a & b
8	0	0	0
1	0	1	0
^	1	0	0
~	1	1	1

14 & 21 (AND)

$$\begin{array}{r}
 14 \Rightarrow 01110 \\
 21 \Rightarrow 10101 \\
 \hline
 00100 \\
 \Rightarrow 4
 \end{array}$$

a	b	a b
0	0	0
0	1	1
1	0	1
1	1	1

14 | 21 (OR)

$$\begin{array}{r}
 14 \Rightarrow 01110 \\
 21 \Rightarrow 10101 \\
 \hline
 11111 \\
 \Rightarrow 31
 \end{array}$$

a	b	a ^ b
0	0	0
0	1	1
1	0	1
1	1	0

14 ^ 21 (XOR)

$$\begin{array}{r}
 14 \Rightarrow 01110 \\
 21 \Rightarrow 10101 \\
 \hline
 11011 \\
 \Rightarrow 27
 \end{array}$$

Lec 4

(~) Bitwise NOT

-ve number

$$\sim(13) \Rightarrow \sim(01101) \Rightarrow 10010$$

$$13 \Rightarrow 01101$$

Sign bit

+ve $\rightarrow 0$

-ve $\rightarrow 1$

negative number always store in 2's complement form.

(Reverse)

$$0010 \rightarrow 1's \text{ complement} \rightarrow 1101$$

$$1101 \rightarrow 2's \text{ complement} \rightarrow 1101 + 1$$

(+1)

$$\therefore \sim(13) \Rightarrow -14$$

$$\Rightarrow 14$$

$$\sim(10) \Rightarrow \sim(01010) \Rightarrow 10101$$

$$10 \Rightarrow 01010$$

+ve

$$0101 \rightarrow 1's \text{ Comp.} \rightarrow 1010$$

$$1010 \rightarrow 2's \text{ comp.} \rightarrow 1010 + 1$$

$$\therefore \sim(10) \Rightarrow -11$$

$$\sim(-14) \Rightarrow \sim(11110) \Rightarrow 00001 \Rightarrow \sim(10010) \Rightarrow 01101$$

$$14 \Rightarrow 11110$$

$$0001 \rightarrow 1's \rightarrow 1110$$

$$1110 \rightarrow 1's \rightarrow 0001$$

$$0001 \rightarrow 2's \rightarrow 0001 + 1$$

$$\therefore \sim(-14) = 13$$

Formula

$$\sim(n) = -(n+1)$$

$$\sim(-14) = -(-14+1)$$

$$= -(-13)$$

$$= 13$$

Shift Operator

Left Shift Operator ($\ll$)

(Multiply by 2)

$$n \ll k = n * 2^k$$

$1 \rightarrow 1$
 $10 \rightarrow 2$
 $100 \rightarrow 4$
 $1000 \rightarrow 8$
 $10000 \rightarrow 16$

$14 \ll 3$

$\Rightarrow 14 * 2^3$

$\Rightarrow 112$

(Divide by 2)

Right Shift Operator ($\gg$)

$10000 \rightarrow 16$

$1000 \rightarrow 8$

$100 \rightarrow 4$

$10 \rightarrow 2$

$1 \rightarrow 1$

$0 \rightarrow 0$

$$n \gg k = n // 2^k$$

$1110 \rightarrow 111$
 $14 \rightarrow 7$
 3

$14 \gg 2$

$\Rightarrow 14 // 2^2$

$\Rightarrow 14 // 4$

$\Rightarrow 3.5 \Rightarrow 3$

membership operator

in, not in

players = ["Sachin", "Kohli", "Dhoni"]

print("Sachin" in players) $\Rightarrow$ True

print("Kohli" not in players) $\Rightarrow$ False.

print("Rohit" in players) $\Rightarrow$ False.

str1 = "Kohli is best ODI player"

print("best" in str1) $\Rightarrow$ True

print("est" in str1) $\Rightarrow$ True

print("tse" in str1) $\Rightarrow$ False.

Identity operator

is, is not

$l_1 \Rightarrow [1, 2, 3]$

$l_2 \Rightarrow [1, 2, 3]$

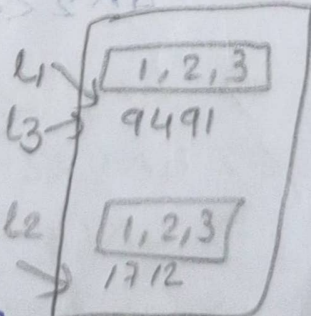
$l_3 \Rightarrow l_1$

print(l1 is l2) $\Rightarrow$ False

print(l1 is not l2) $\Rightarrow$ True

print(l1 is l3) $\Rightarrow$ True

print(l2 is l3) $\Rightarrow$ False.



$\Rightarrow$ Check 2 variables pointing out or same address.

Operator Precedence

$$\Rightarrow 5 + 12 * 7 / 2 - 10$$

$$5 + 84 / 2 - 10$$

$$5 + 42.0 - 10$$

$$47.0 - 10$$

$$37.0$$

$$\bullet \text{ point } (2^{**} 3^{**} 2)$$

$$\Rightarrow 2^{**} 9$$

$$\Rightarrow 512$$

Precedence

1) Parenthesis

2) $**$ (Exponent)

3) $*$, $/$, $\%$, $//$

4) $+$, $-$

Associativity

Left to Right

Right to Left

Left to Right

Left to Right

precedence	Operator	Associative
1	Parenthesis	Left to Right
2	Exponent	Right to Left
3	Unary ($++$, $+$, $-$)	right to left
4	$*$, $/$, $\%$, $//$	left to right
5	$+$, $-$	left to right
6	$<<$, $>>$	left to right
7	$\&$	left to right
8	$\wedge$	left to right
9	$!$	left to right
10	$=$, $!=$, $<$, $>$, $<=$, $>=$, in, not in, is, is not	left to right
11	NOT	Right to Left
12	AND	left to Right
13	OR	left to Right

$$\bullet x = 8$$

$$\text{point } (x >> 2 << 1)$$

$$\Rightarrow (8 >> 2 << 1) \quad \begin{matrix} k=1 \\ n=2 \end{matrix}$$

$$8 // 2^k$$

$$2 // 1 \quad 2 << 1$$

$$8 // 4$$

$$\Rightarrow 2 \quad n // 2^k$$

$$\Rightarrow 2$$

$$2 * 2^1$$

$$\Rightarrow 4$$

$$\bullet 4 \wedge 2 << 3 + 4 * 8 // 2 * 4$$

$$\Rightarrow 4 \wedge 2 << 3 + 2$$

$$\Rightarrow 4 \wedge 2 << 5$$

$$\Rightarrow 4 \wedge 64$$

$$\Rightarrow 4 \Rightarrow 0000100$$

$$64 \Rightarrow 10000000$$

$$01110111$$

$$10000000$$

$$(68)$$

• point $(10 > 7 == 7 < 12)$
 comparison operators follow chain Rule.

$$\begin{array}{ccc} 10 > 7 & \text{and} & 7 == 7 & \text{and} & 7 < 12 & \Rightarrow & \text{True} \\ \underline{T} & & \underline{T} & & \underline{T} & & \end{array}$$

• $(10 > 7 == 8 < 12)$

$$\Rightarrow \begin{array}{ccc} 10 > 7 & \text{and} & 7 == 8 & \text{and} & 8 < 12 & \Rightarrow & \text{False (Short Circuit)} \\ \underline{T} & & \underline{F} & & & & \end{array}$$

• $(10 > 7) == 8 < 12$

$$\text{True} == 8 < 12$$

$$\text{True} == 8 \text{ and } 8 < 12 \Rightarrow \text{False (Short Circuit)}$$

False

• $10 > (7 == 8) < 12$

$$10 > \text{False} < 12$$

$$10 > \text{False} \text{ and } \text{False} < 12 \Rightarrow \text{True}$$

False == 0 / False

True: 1 < 0

• $10 > 7 == (8 < 12)$

True

$$10 > 7 == \text{True}$$

$$10 > 7 \text{ and } 7 == \text{True} \Rightarrow \text{True}$$

True

False

True
False

Unicode Representation

Ascii Code (7bits)

a=97	A=65
b=98	B=66
c=99	C=67
⋮	⋮

• print(ord('A'))
97 → 65

